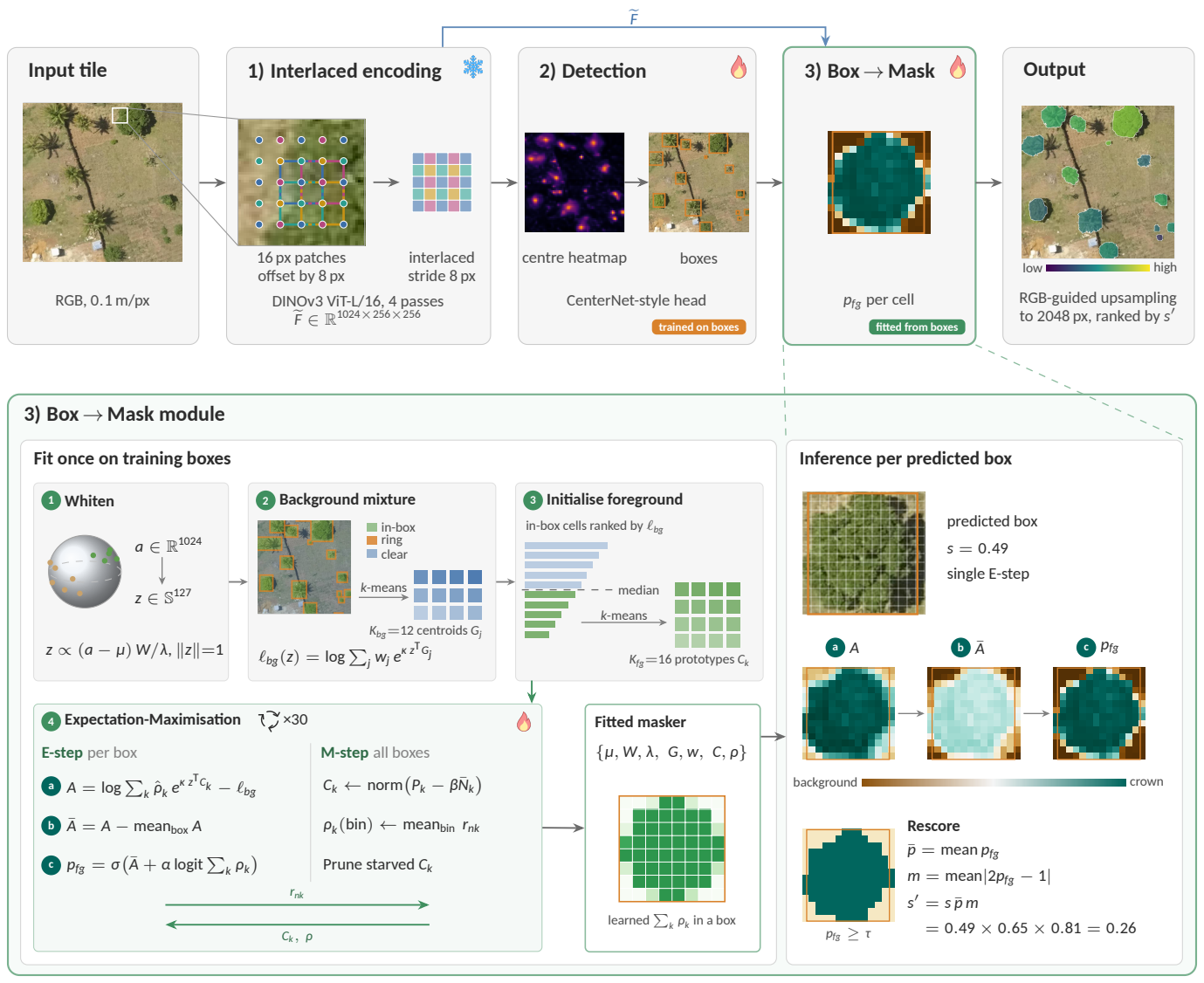

$\widetilde{F}$
Input tile
RGB, 0.1 m/px
1) Interlaced encoding
16 px patches offset by 8 px
interlaced stride 8 px
DINOv3 ViT-L/16, 4 passes
$\widetilde{F} \in \mathbb{R}^{1024 \times 256 \times 256}$
2) Detection
centre heatmap
boxes
CenterNet-style head
trained on boxes
3) Box → Mask
p_{fg} per cell
fitted from boxes
Output
low
high
RGB-guided upsampling to 2048 px, ranked by s'
3) Box → Mask module
Fit once on training boxes
1 Whiten
$a \in \mathbb{R}^{1024}$
$z \in \mathbb{S}^{127}$
$z \propto (a - \mu)\, W/\lambda,\ \|z\|=1$
2 Background mixture
in-box
ring
clear
k-means
K_{bg}=12 centroids G_j
$\ell_{bg}(z) = \log \sum_j w_j\, e^{\kappa\, z^T G_j}$
3 Initialise foreground
in-box cells ranked by ℓ_{bg}
median
k-means
K_{fg}=16 prototypes C_k
4 Expectation-Maximisation
×30
E-step per box
a $A = \log \sum_k \hat{\rho}_k\, e^{\kappa\, z^T C_k} - \ell_{bg}$
b $\bar{A} = A - \text{mean}_{\text{box}}\, A$
c $p_{fg} = \sigma(\bar{A} + \alpha\, \text{logit} \sum_k \rho_k)$
M-step all boxes
$C_k \leftarrow \text{norm}(P_k - \beta \bar{N}_k)$
$\rho_k(\text{bin}) \leftarrow \text{mean}_{\text{bin}}\ r_{nk}$
Prune starved C_k
r_{nk}
$C_k,\ \rho$
Fitted masker
$\{\mu, W, \lambda,\ G, w,\ C, \rho\}$
learned $\sum_k \rho_k$ in a box
Inference per predicted box
predicted box
$s = 0.49$
single E-step
a A
b $\bar{A}$
c p_{fg}
background
crown
$p_{fg} \geq \tau$
Rescore
$\bar{p} = \text{mean}\, p_{fg}$
$m = \text{mean}|2p_{fg} - 1|$
$s' = s\, \bar{p}\, m$
$= 0.49 \times 0.65 \times 0.81 = 0.26$